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Homework 2

1. What is Hardware Trojan? What hardware Trojans can do? Which applications are likely to be targets for attackers?

A hardware trojan is a malicious and intentional modification of an integrated circuit or hardware design that alters its intended functionality, leaks sensitive information or degrades the systems performance. Hardware Trojans can be inserted during different stages of the design and fabrication lifecycle, particularly in globally distributed and potentially untrusted supply chains (Bhunja & Tehranipoor, 2019). They typically consist of two components, a trigger mechanism and the payload. The damage they can inflict is quite devastating and some examples of what they can do are leak cryptographic keys and confidential data, modify system outputs, degrade performance, create denial-of-service conditions and insert backdoors for future exploitation. With this wide variety of abilities, the target range is also large being military and defense systems, cryptographic hardware, financial and banking infrastructures, IoT systems, embedded systems and critical infrastructures. These systems are all attractive because they contain and handle sensitive information or are in control of safety-critical operations.

2. Explain VLSI Design Cycle in brief.

The Very Large-Scale Integration (VLSI) design cycle follows a structured flow to transform system specifications into a fabricated chip. Below is a brief rundown of the cycle.

1. **Specification** - Defines functional and performance requirements
2. **Architectural Design** - Develops high-level system architecture
3. **Register Transfer Level (RTL) Design** - Describes hardware behavior using HDL
4. **Functional Verification** - Ensure RTL correctness via simulation
5. **Logic Synthesis** - Convert the RTL into gate-level netlist
6. **Physical Design** - Placement, routing and clock tree synthesis
7. **Design Rule Checking (DRC) and Verification** - Validation of layout compliance
8. **Fabrication** - Send design out to fabrication.
9. **Testing and Packing** - Post-silicon validation and defect testing

Security risks may be introduced at multiple stages of this design cycle but in most cases, it is during IP integration and fabrication (Bhunias & Tehranipoor, 2019).

3. Explain Logic Testing & Side Channel Analysis with their pros and Cons.

Logic Testing is the process of detecting hardware Trojans by applying test patterns and then observing the output responses for anomalies. The advantage of this is that it is compatible with existing design for test infrastructure. Also, it is effective for Trojans that alter functional outputs and it requires relatively low additional hardware overhead. However, it is ineffective for Trojans with rare triggers, hard to detect or dormant Trojans and it requires extensive test pattern generation. Side-Channel Analysis (SCA) in comparison measures indirect physical effects such as power consumption, electromagnetic emissions and timing variations to detect Trojans. With this process it can detect stealthy trojans not visible in logic outputs and it is effective for small

structural modifications. But it is sensitive to process variations and noise, requiring precise measurement equipment. In addition, it can produce high false positives due to manufacturing variations. So, SCA's power comes from the fact that Trojans often introduce measurable physical deviations, even if it is functionally dormant (Bhunia & Tehranipoor, 2019).

4.a. What are the Approaches of Pre-synthesis? Explain them in brief.

Pre-synthesis approaches aim to prevent Trojan insertion before the design is synthesized into gates. The goal is to identify malicious modifications at the RTL or architectural level, where the corrections are less costly and manageable. One important approach is Design Security Rule Check (DSerc), which uses automated rule-checking frameworks to identify violations of security constraints within the design. Another technique is Information Flow Tracking (IFT), which analyzes how sensitive data propagates through the system to ensure that confidential information does not reach unauthorized outputs. Additionally, code review and formal verification techniques are employed to examine the RTL descriptions for malicious alterations or unintended behaviors. IP validation and authentication mechanisms are also critical when integrating third-party intellectual property cores. These techniques verify the integrity and authenticity of externally sourced IP before incorporation into the system. Collectively, pre-synthesis approaches emphasize early detection and prevention, which reduces the risk and cost associated with post fabrication discovery of hardware trojans (Bhunia & Tehranipoor, 2019).

4.b. Discuss the Classification of Trojan Detection Approaches.

Hardware Trojan detection approaches can be classified into pre-silicon detection, post-silicon detection, run-time monitoring and prevention-based methods. The pre-silicon detection techniques aim to identify Trojans before fabrication through methods such as formal

verification, RTL logic testing and information flow analysis. This is to ensure that the design satisfies the security properties prior to tape-out. In contrast, post-silicon detection occurs after the fabrication and relies on techniques such as side-channel analysis, functional testing and physical inspection. Side-channel analysis was previously discussed above and functional testing evaluates abnormal circuit behavior under test patterns, while physical inspection techniques may involve imaging or reverse engineering. Incorporation of on-chip sensors and security monitors to detect anomalous behavior during operation is done during run-time monitoring, allowing for dynamic identification of Trojan activation. While prevention-based approaches rely on design-for-trust (DfT), logic obfuscation and hardware watermarking to make Trojan insertion more difficult or detectable. Each classification presents its own trade-offs in terms of cost, detection accuracy, implementation complexity and scalability. Since no single technique guarantees complete coverage, a layered defense strategy combining multiple approaches is recommended for ensuring proper security (Bhunia & Tehranipoor, 2019).

References

Bhunia, S., & Tehranipoor, M. H. (2019). *Hardware security: A hands-on learning approach*. Morgan Kaufmann.